

Test and Measurement of RF/Wireless Systems

More and more devices today have RF wireless circuitry inside, whether it is something obvious like a cell phone or something as innocuous as a new car key. RF/wireless testing for these devices is very vital as the actual working of the setup cannot be validated without testing the system that has been developed and deployed



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Wireless communication can be defined as the transfer of information between two or more points that are not connected by an electrical conductor. Radio is the most often used electromagnetic wireless telecommunication standard for wireless communication. It allows long-range communications that are close to impossible or impractical to implement with the use of wires. This term is most commonly used in the telecommunications industry to refer to telecommunication systems that use some form of energy (radio waves, acoustic energy, etc) to transfer information without the use of wires. The best part is that this sort of communication can be used over both long and short range.

With the exploding use of cell phones, wireless personal digital assistants (PDAs), Bluetooth headsets, RFID tags, wireless medical devices and ZigBee sensors to mention a few,

the RF device market is booming. These wireless devices communicate over the open airways, and their operation depends upon careful consideration and design of their operating frequency, power level, signalling format and other technical issues.

Need for standardised test and measurement

Since RF devices are wireless, design and testing presents an array of new and varied challenges to test engineers.

Madhukar Tripathi, regional manager, Anritsu India, says, "The need for RF/wireless testing is very vital as the actual working of the setup cannot be verified without testing the system that has been developed and deployed."

Stephen Hire, general manager, Aeroflex Asia-India, adds, "The decision on which parameters to test depends to a large extent on the technology and the stage of the life-cycle that the engineers are involved in."

In the context of LTE and the latest WLAN standard, 802.11ac, he shares, "For RF engineers, digital technology puts a lot of stress on the receiver chain. So measurements such as error vector magnitude are critical. Bandwidths are also ever-increasing (multiple blocks of 20 MHz for LTE-Advanced and 160 MHz for 802.11ac), which makes designing individual components more challenging to ensure even performance over a wide frequency band. For protocol engineers, standards are growing complex day-by-day and everything

from channel coding to signalling, messaging and error handling needs to be checked to ensure they are absolutely in line with specifications. System test engineers then need to check real-world performance including resilience to fading, corrupted messages and, of course, that the systems are delivering the very high data rates that are the main selling points for these new standards."

With the explosion in wireless products, especially mobile devices, the need for standardised test and measurement procedures is also on the rise.

Challenges for RF/wireless engineers

There are significant issues and challenges that device manufacturers face in testing their devices to specifications. In some instances, they may want to test beyond specifications to ascertain margins in the electrical performance of a specific device.

J.K. Baldua, director-technical, Scientech Technologies, opines, "Detecting low-level signals buried in plenty of noise and segregating each frequency carrying the signal separately in a bunch of frequency spectrum has been always challenging, because this needs much higher resolution of bandwidth."

Madhukar Tripathi says, "Since technological developments are always on the run, the root-level testing of any RF/microwave setup at the field is now becoming very important and very vital. For example, if the user has developed something for which over-the-air communication is a major application and has deployed the system in field, it becomes very important to verify the system's performance over the air. The re-